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Hospital Emergency Resource Allocation System Using OpenVINO: A Machine Learning-Based Approach to Inference Optimization and Resource Planning

BY:

26MCS0095 Radhika M

26MCS0071 Pavithra S

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

VELLORE INSTITUTE OF TECHNOLOGY, VELLORE-600127

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2.Abstract with problem statement and result achieved:

Hospital emergency departments frequently experience unpredictable patient arrivals, overcrowding and inefficient utilization of limited resources such as emergency beds, medical staff, patient monitors and oxygen units. Traditional resource-planning approaches may not respond efficiently to changing patient demand. Machine learning can assist in analysing emergency department data and predicting patient acuity, while inference optimization can help evaluate the feasibility of deploying such models efficiently.

This project presents a Hospital Emergency Resource Allocation System using machine learning and Intel OpenVINO. The implemented prototype uses emergency department data containing patient demographics, triage information, vital signs, diagnoses and visit-disposition details. Data inspection and preprocessing are performed before training classification models. Random Forest is evaluated as a baseline model, followed by a Logistic Regression experiment. The trained Logistic Regression classifier is converted into ONNX format and executed using OpenVINO on the CPU.

The experimental results recorded a baseline Random Forest accuracy of 46.15% and balanced accuracy of 39.34%. Logistic Regression achieved an accuracy of 51.92% on the reported test split. OpenVINO produced predictions identical to those of the original scikit-learn classifier for all 207 labelled records in the reported validation experiment. Repeated classifier-only benchmarks recorded speed ratios of approximately 1.72× and 1.83× in separate experiments. However, end-to-end measurements showed that preprocessing and execution overhead can reduce or eliminate these gains.

The project demonstrates the feasibility of integrating machine-learning inference optimization into an emergency department decision-support prototype. The current results are preliminary because the available dataset is small and imbalanced. The system is intended to support operational planning rather than replace clinical judgement. Further work is required to validate resource-demand predictions, evaluate allocation strategies and test performance on additional hardware and independent datasets.

Keywords: *Emergency Department, Resource Allocation, Machine Learning, Logistic Regression, Random Forest, OpenVINO, ONNX, Inference Latency*

2.1Problem statement

Emergency departments must make operational decisions under changing patient demand and limited resource availability. Inefficient resource planning can contribute to delays and overcrowding. This project investigates whether machine-learning classification and optimized inference can support emergency department decision-making while measuring the trade-offs between prediction quality, computational cost and execution time.

The current implementation focuses on model training, model conversion, prediction agreement, performance benchmarking and a basic resource inventory. A fully validated, automated clinical resource-allocation system has not yet been demonstrated.

Result

Experiment	*Recorded result*
Random Forest baseline accuracy	*46.15%*
Random Forest balanced accuracy	*39.34%*
Logistic Regression accuracy	*51.92%*
OpenVINO/scikit-learn prediction agreement	*100% on 207 records*
Classifier-only speed ratio, repeated batch benchmark	*1.72x*
Classifier-only speed ratio, single-record benchmark	*1.83x*
End to end speed ration in the third reported	*Approximately 1.01x*

3.Introduction:

Emergency departments are critical components of healthcare systems because they provide immediate medical attention to patients with varying levels of urgency. Effective emergency department management requires timely patient assessment, appropriate prioritization and efficient utilization of available resources. However, unpredictable patient arrivals, overcrowding and limited staff and equipment can create operational challenges.

Machine learning provides methods for identifying patterns in historical healthcare data and supporting predictions about patient outcomes, workload and emergency department demand. Zlotnik et al. (2015) investigated machine-learning techniques for emergency department visit forecasting and dynamic nursing staff allocation. Their work demonstrates the relevance of predictive analytics to healthcare workforce planning. Similarly, Mizan and Taghipour (2022) investigated the integration of machine-learning prediction and mathematical optimization for medical resource allocation.

Although predictive models can provide useful information, their practical deployment also depends on computational efficiency. Intel OpenVINO provides an inference toolkit that supports optimized execution of compatible trained models on supported hardware. A trained model can be converted into a suitable representation and evaluated to determine whether inference optimization improves execution performance without changing the model's predictions.

The proposed project, Hospital Emergency Resource Allocation System Using OpenVINO, combines emergency department data analysis, machine-learning classification, optimized inference and basic resource inventory tracking. The implementation evaluates Random Forest and Logistic Regression experiments and deploys the Logistic Regression classifier through ONNX and OpenVINO. Its performance is examined using prediction agreement, classifier-only latency and end-to-end latency.

The current prototype establishes a foundation for machine-learning-assisted emergency department decision support. The available experiments focus on triage classification and inference optimization; they do not yet establish the clinical effectiveness of automated resource allocation. Therefore, the system is intended to support, rather than replace, decisions made by qualified healthcare professionals.

Project workflow:

Emergency department dataset → Data inspection and preprocessing → Model training and evaluation → ONNX conversion → OpenVINO CPU inference → Performance benchmarking → Resource inventory demonstration.

Literature citations

1. Zlotnik, A.,et al. *Emergency department visit forecasting and dynamic nursing staff allocation using machine learning techniques with readily available open-source software*. Computers, Informatics, Nursing, 33(8), 368–377.
2. Mizan, T., & Taghipour, S. (2022). *Medical resource allocation planning by integrating machine learning and optimization models*. Artificial Intelligence in Medicine, 134, 102430.
3. El Ariss, A. B., et al. (2024). *Development and validation of a machine learning framework for improved resource allocation in the emergency department*. The American Journal of Emergency Medicine, 84, 141–148.

4. Motivation & Feasibility:

4.1.Technical Feasibility :

The project is technically feasible because publicly available healthcare datasets can be used for model development. Machine-learning algorithms such as Random Forest, XGBoost, LightGBM, Logistic Regression and neural networks have already demonstrated usefulness in emergency-department prediction tasks.

OpenVINO provides tools for converting, optimizing and deploying trained models and provides benchmarking facilities for measuring inference latency and throughput.

The proposed system can therefore be implemented using:

- Python
- Pandas and NumPy
- Scikit-learn
- PyTorch
- OpenVINO
- Streamlit

The prototype successfully executes a converted Logistic Regression classifier through OpenVINO on the CPU. The reported validation experiment achieved 100% prediction agreement with the original scikit-learn classifier on 207 labelled records. Repeated benchmarks also demonstrate that classifier-only performance and complete-pipeline performance must be evaluated separately.

4.2.Economic Feasibility:

The prototype uses software tools and libraries that are available for development without requiring a dedicated commercial AI server. Development and initial testing can be performed on an existing computer.

If optimized inference operates efficiently on existing hardware, it may reduce the need for additional computational infrastructure. However, the current experiments do not establish financial savings or quantify the cost of deploying the system in a hospital. Such benefits would require a separate cost analysis.

4.3. Social Feasibility:

Emergency department overcrowding and inefficient resource utilization can affect patients, clinicians and hospital administrators. A reliable decision-support system could help staff understand patient acuity patterns and resource availability, potentially supporting better operational planning.

The current prototype is not clinically validated and should not make autonomous treatment or triage decisions. Healthcare professionals must retain responsibility for patient care, and the system's outputs should be evaluated for reliability and potential bias before any clinical use.

4.4. Environmental Feasibility:

Inference optimization and local execution may reduce unnecessary transmission of patient information to remote services. Efficient use of existing hardware could also reduce the need for additional computing resources.However, environmental benefits have not been measured in this project. Establishing them would require measurements of energy consumption, hardware utilization and the computational cost of alternative deployment approaches.

4.5.Political/Regulatory Feasibility:

Healthcare software must respect applicable patient privacy, data-protection, cybersecurity and medical-device requirements. The prototype should use appropriately authorized, de-identified data and restrict access to any sensitive information.Before real-world deployment, the system would require appropriate institutional approval, security assessment, clinical evaluation and compliance review. OpenVINO execution alone does not establish regulatory approval or clinical safety.

4.6.Demographic Feasibility :

Emergency departments serve patients from different age groups and demographic backgrounds. Patient age, sex and other relevant characteristics may influence the distribution of emergency presentations and model predictions.

The dataset should therefore be examined for demographic representation and missingness. Where sample sizes permit, performance should be evaluated across demographic subgroups to identify possible disparities. The current small labelled sample is insufficient to establish fairness or generalizability across populations.

Individual Member Contribution

Member	Contribution
Radhika M	**Literature Survey, Dataset Collection,Model Training**
Pavithra S	**Fine-tuning, and OpenVINO IR (Intermediate Representation) Conversion.**

5.***Project Plan:***

Activity	**Duration**	**Planned Period**
Literature Survey & Requirement Analysis	7days	Aug 19 – Aug 25
Dataset Collection & Preprocessing	10 days	Aug 26 -Sep 4
Patient & Crowd Detection Model	10 days	Sep 5 -Sep 14
Triage Severity Classification	7 days	Sep 15-Sep 21
Resource Inventory & State Tracking	5days	Sep 22-Sep 26
Resource Allocation Algorithm & Logic	10 days	Sep 27- Oct 6
OpenVINO Optimization	7days	Oct 7-Oct 13
System Integration	7 days	Oct 14 - Oct 20
Testing & Debugging	7 days	Oct 21 - Oct 27
Performance Evaluation	5 days	Oct 28 -Nov 1
Final Documentation & Report Preparation	7 days	Nov 2 -Nov 8

6. ***Literature Review:***

No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Environment	Future Work
1	Modeling daily patient arrivals at Emergency Department and quantifying the relative importance of contributing variables using artificial neural network	Decision support systems(Science Direct) https://tinyurl.com/363eep36	Predicting daily ED patient arrivals for better resource planning.	Artificial Neural Network , Multiple linear Regression	ANN performed better than regression models in arrival prediction; temporal and influenza factors were important.	One year of data from a local ED.	Good prediction identifies important variables.	Single ED; limited dataset duration.	Emergency Department	Test on multiple hospitals and incorporate real-time data
2	Emergency Department Visit Forecasting and Dynamic Nursing Staff Allocation Using Machine Learning Techniques With Readily Available Open-Source Software	Computers, Informatics, Nursing, 2015 https://tinyurl.com/7z5yvcf5	Forecasting ED visits and dynamically allocating nurses.	SVR, Time series model, regression M5P (model tree with linear regression on leaf nodes)	SVR and M5P outperformed the baseline; dynamic allocation could substantially reduce understaffing.	Data from 1100 bed specialized care hospital involving approx 553,000 patients.	Directly connects forecasting with staffing.	Focused mainly on nursing workforce	Hospital emergency department.	Integrate more resources and modern deep learning methods.

No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
3	Multi-objective simulation optimization for medical capacity allocation in emergency department	Journal of simulation https://doi.org/10.1057/jos.2014.39 Pubmed	Optimizing doctors, nurses, laboratory staff and equipment under capacity constraints.	NSPSO, MOCBA, simulation optimization (Multi-Objec tive Optimal Computing Budget Allocation)	Generated optimal resource allocation solutions while considering patient stay and resource cost.	ED operation, simulation data from Taiwan.	Handles multiple objectives & constraint	Simulation dependent computational complexity.	Emergency Department	Combine simulation optimization with ML-based demand prediction
4	Chaotic genetic algorithm and Adaboost ensemble metamodeling approach for optimum resource planning in emergency departments	PubMed https://doi.org/10.1016/j.artmed.2017.10.002	Reducing ED overcrowding and length of stay through optimum resource allocation.	Agent-based simulation, ANFIS, FFNN, RNN, AdaBoost, Genetic Algorithm	Reduced average length of stay by 14%; metamodel improved MAPE by 26.6% over individual models.	Public ED case-study /simulation data	Combines ML and optimization.	Computationally expensive; simulation assumptions.	Public Emergency Department	Develop faster real-time optimization and integrate live hospital data.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
5	Predicting hospital admission at emergency department triage using machine learning	Plos one Science Direct https://doi.org/10.1371/journal.pone.0201016	Predicting hospital admission during ED triage.	Logistic Regression, XGBoost, Deep Neural Networks	XGBoost achieved **AUC 0.91** using triage and historical information	Adult ED visits from one academic and two community EDs; 2014–2017.	High predictive performance; useful for bed planning.	Dataset was from one healthcare system.	Emergency Department Triage	External validation and integration with bed/resource allocation.
6	Emergency department triage prediction of clinical outcomes using machine learning models	Springer https://tinyurl.com/bdhrxa36	Predicting critical care and hospitalization outcomes at triage.	Lasso, Random Forest, Gradient Boosting, DNN	DNN achieved AUC 0.86 for critical care and 0.82 for hospitalization, outperforming conventional ESI.	135,470 adult ED visits	Large dataset; compares ML with conventional triage.	Requires external validation.	ED triage	Prospective validation and clinical deployment.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
7	Forecasting emergency department overcrowding: A deep learning framework	Science Direct https://doi.org/10.1016/j.chaos.2020.110247	Predicting ED overcrowding and patient flow problems.	VAE, RNN, LSTM, BiLSTM, ConvLSTM, GRU, CNN	Eight deep-learning models were compared; VAE showed superior performance.	ED data from CHRU Lille, France.	Compares several DL architectures. .	Single healthcare setting; generalizability issue.	Emergency Department Triage	Multicenter validation and real-time deployment.
8	Prediction of emergency department resource requirements during triage: An application of current natural language processing techniques	PubMed https://doi.org/10.1002/emp2.12253	Predicting the number of resources required by a patient during triage.	NLP + Neural Network + clinical variables	Accuracy 66.5% and macro-F1 0.601 for predicting resource categories.	226,317 ED encounters from three academic EDs.	Directly predicts resource requirements; uses triage notes.	Less accurate for high-resource patients.	ED Triage	Improve NLP, include more clinical variables and deploy prospectively.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
9	Towards a real-time prediction of waiting times in emergency departments: A comparative analysis of machine learning techniques	Science Direct https://doi.org/10.1016/j.ijforecast.2021.10.006	Predicting ED waiting times to support operational decisions and overcrowding management.	Random Forest, SVR, ANN, Ensemble	Ensemble method was most effective; Random Forest provided a good accuracy computation trade-off.	Two real ED datasets: 40,504 and 60,230 presentations.	Real-world data; supports real-time monitoring.	Waiting time prediction does not directly allocate resources.	Two Hospital EDs	Integrate waiting-time prediction with automatic resource allocation.
10	Machine Learning Methods for Predicting Patient-Level Emergency Department Workload	Science Direct https://tinyurl.com/pzrr268a	Predicting workload generated by individual ED patients for physician load balancing.	Linear Regression, Gradient Boosted Tree, Neural Networks	Neural network using text achieved lowest MAE of 0.78 wRVUs.	204,064 adult ED visits	Directly measures patient level workload.	Single-center study; wRVU is an indirect workload measure.	Urban Academic ED	Multicenter validation and real-time staff balancing.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
11	Multisite evaluation of prediction models for emergency department crowding before and during the COVID-19 pandemic	PubMed https://doi.org/10.1093/jamia/ocac214	Evaluating whether ED crowding models remain reliable under temporal and geographical changes.	Random Forest	Evaluated model performance under COVID related temporal drift and hospital drift.	Four datasets from academic and rural hospitals; pre COVID and COVID periods.	Studies robustness and model drift.	Performance can degrade under distribution shift.	Multiple Emergency Department	Continuous monitoring, recalibration and external validation.
12	Medical resource allocation planning by integrating machine learning and optimization models	Science Direct https://doi.org/10.1016/j.artmed.2022.102430	Allocating medical resources while reducing workload and patient waiting time.	ML forecasting + Mathematical Optimization	Reduced radiologist workload by 25% and patient waiting time by 8.17%.	Health care patient arrival and radiology workload data.	Strong combination of prediction and optimization.	Focuses mainly on radiology resources.	Healthcare Resource Planning	Extend to multiple hospital resources and real-time allocation.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
13	Time-series Machine Learning Models to Support Emergency Department Operational Planning	PubMed https://pmc.ncbi.nlm.nih.gov/articles/PMC12919617/	Forecasting future ED demand and admissions.	Time-series models, ML, ensemble/ stacked prediction	One-day-ahead forecasts achieved MAPE around 6.8% and 8.6% for two hospitals.	8 years of electronic admissions data from two London hospitals.	Combines traditional and ML approach	Linear models sometimes outperformed complex ML models.	UK Hospital EDs	Apply framework to more hospitals and operational indicators.
14	Forecasting emergency department occupancy with advanced machine learning models and multivariable input	Science Direct https://doi.org/10.1016/j.ijforecast.2023.12.002	Predicting hourly, 4-hour, 8-hour and daily ED admissions.	SARIMA, Exponential Smoothing, RNN, XGBoost, Ensemble Learning	RNN and XGBoost produced the best hourly forecasts; XGBoost was computationally efficient.	10 years (2009–2018) of admissions from a Portuguese hospital	Multiple prediction horizons; compares several models.	Single hospital; limited generalization.	Portuguese Hospital ED	Multihospital testing and integration with staff scheduling

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
15	Forecasting emergency department occupancy with advanced machine learning models and multivariable input	Science Direct https://doi.org/10.1016/j.ijforecast.2023.12.002	Forecasting ED occupancy up to 24 hours ahead.	DeepAR, N-BEATS Temporal Fusion Transformer, LightGBM	Advanced ML models outperformed traditional benchmarks by up to 15%.	EHR data plus hospital-bed availability, traffic and weather data.	Uses multiple external variables and advanced models.	More complex models require greater computational resources.	Large Combined ED	Real-time deployment and integration with allocation systems.
16	Development and validation of a machine learning framework for improved resource allocation in the emergency department	Science Direct https://doi.org/10.1016/j.ajem.2024.07.040	Predicting individual patient resource requirements during ED triage.	Azure AutoML, 144 Voting Ensemble models	391,472 ED visits; average AUC 0.82 and accuracy 0.76 across 144 resource targets.	MIMIC-IV ED, 391,472 visits.	Most directly related to the proposed project; predicts many resource types.	Retrospective data requires external validation.	Academic Hospital ED	Expand datasets, increase resource targets and move toward real-time deployment.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
17	Improving triage performance in emergency departments using machine learning and natural language processing: a systematic review	Springer https://link.springer.com/article/10.1186/s12873-024-01135-2	Reviewing ML/NLP approaches for improving ED triage.	Systematic Review; ML and NLP analysis	Reviewed 60 studies and 57 ML algorithms XGBoost, Gradient Boosting and DNN showed strong performance.	60 published studies from PubMed, Scopus, Web of Science, IEEE Xplore and ACM.	Broad review; identifies common predictors and methods.	High risk of bias in many models; limited XAI.	Emergency Department Triage	More explainable, unbiased and externally validated models.
18	Enhanced forecasting of emergency department patient arrivals using feature engineering approach and machine learning	Springer https://tinyurl.com/a9v2wd79	Improving ED arrival forecasting for better staff and resource allocation.	XGBoost, LightGBM, SVM, Feature Engineering	XGBoost was best; MAPE ranged from 5.03%–14.1% across datasets.	11 EDs across 3 countries, 2014 - 2016	Multicenter and multinational dataset; feature engineering improves prediction.	Mainly predicts arrivals rather than actual resource allocation.	Multiple EDs	Integrate arrival forecasts with automated allocation and optimization.

Ref No	Title of the paper	Journal Name	Attack problem	Method	Result	Dataset	Pros	Cons	Env	Future Work
19	Predictive modeling of hospital emergency department demand using artificial intelligence: A systematic review	Science Direct https://doi.org/10.1016/j.ijmedinf.2025.106215	Reviewing AI/ML methods for ED demand forecasting and identifying research gaps.	PRISMA Systematic Review; ML/DL comparison	ML,DL generally outperform classical methods; external validation and explainability remain limited.	11 eligible studies published between 2019 and July 2025.	Very recent review; directly identifies research gaps.	Review does not implement a new allocation system.	Hospital ED Demand Forecasting	Multicenter validation, explainable AI and hybrid forecasting.
20	An Artificial Intelligence-Based Framework for Predicting Emergency Department Overcrowding: Development and Evaluation Study	PubMed https://doi.org/10.2196/73960	Predicting imminent ED crowding in real time.	LSTM Neural Network	One-hour-ahead prediction achieved AUC 0.84, compared with 0.73 for modified NEDOCS.	Single-center EHR and workforce data.	Real-time prediction; directly supports ED workflow.	Single center retrospective study; requires external validation	Emergency Department	Multicenter validation and prospective real-time implementation

7.Research Gaps

1: Prediction is often separated from allocation

Existing research frequently focuses either on predicting emergency department demand and allocating healthcare resources or on optimizing the execution of trained machine-learning models. These two objectives are not always evaluated together in a single experimental workflow

Research gap: There is a need to investigate how inference optimization can be incorporated into an emergency department decision-support pipeline and how computational improvements relate to the overall resource-planning process.

Proposed approach: This project combines emergency department data processing, machine-learning classification, OpenVINO inference and a basic resource inventory demonstration within one prototype.

2: Limited evaluation of end-to-end inference performance

A model that executes faster in isolation does not necessarily make an entire application faster. Data preprocessing, input conversion, model execution and Python-level overhead can affect the total prediction time.

Research gap: Performance evaluations based only on model execution may not accurately represent the latency experienced by an application using the complete inference pipeline.

Proposed approach: The project evaluates both classifier-only latency and end-to-end latency. The results show that OpenVINO can improve classifier-only execution time, while the improvement is not consistently reflected in complete-pipeline measurements.

3: Need to evaluate prediction consistency after model conversion

Deploying a trained model through an inference optimization framework requires checking that the converted model produces predictions consistent with the original implementation.

Research gap: Inference speed measurements alone are insufficient to establish that a converted model preserves the original model's predictions.

Proposed approach: The trained Logistic Regression classifier is converted into ONNX format and executed through OpenVINO. The reported validation experiment records 100% prediction agreement across 207 labelled emergency-visit records. This result demonstrates consistency for the tested records but does not independently establish clinical accuracy or generalization to unseen populations.

4: Single-resource focus

Some studies concentrate on only one resource, such as nurses, beds or radiologists.

The proposed system can consider multiple resource categories simultaneously:

- Doctors
- Nurses
- Emergency beds
- Laboratory resources
- Imaging resources
- ICU/critical-care capacity
- Other treatment resources

5: Limited explainability

Hospital decision-makers need to understand why a system predicts high resource demand.

The proposed system should therefore include explainability using methods such as feature importance or SHAP where applicable.

6: Generalizability

Recent reviews highlight insufficient external validation and limited generalizability across hospital. The project can address this experimentally by comparing performance across different temporal splits or available hospital subsets.

7: Performance trade-off between accuracy and inference speed

A highly accurate model may be computationally expensive.

The proposed system will evaluate both:

Prediction quality + inference performance

This makes OpenVINO benchmarking an important component of the project.

8.Overview of Proposed Work :

The proposed Hospital Emergency Resource Allocation System consists of six major stages.

Stage 1: Data Collection

The first module loads the emergency department dataset containing patient visits and associated clinical and administrative information. The project screenshots show separate data tables, including triage information, vital signs, diagnoses, medication information and visit disposition.

The dataset is inspected to understand its structure, identify missing values, examine duplicate records and determine whether records from different tables can be reliably matched.

The inspection in your screenshots reports 222 visit IDs in several tables, with varying overlap across tables. This makes it important to verify record relationships before joining data, as incorrect joins can duplicate patient visits or introduce misleading information.

Potential variables include:

Patient Information

- Age
- Sex
- Used to describe patient demographics.

Vital Signs

- Temperature
- Heart rate
- Respiratory rate
- Oxygen saturation (SpO_2)
- Blood pressure
- Used to represent the patient's physiological condition.

Triage Information

- Triage acuity
- Used as the target variable for predicting the patient's acuity class.

Visit Information

- Visit ID
- Visit disposition
- Used to identify patient visits and describe their outcomes.

Clinical Information

- Diagnosis
- Diagnosis code
- Used to represent recorded medical conditions.

Stage 2: Data Preprocessing and feature preparation

Raw healthcare data cannot always be used directly by a machine-learning model. This module prepares the data for training and inference.

The main operations include:

- Identifying missing and duplicate records.
- Selecting relevant patients and visiting features.
- Preparing numerical features such as temperature, heart rate, respiratory rate and oxygen saturation.
- Encoding categorical features into a format suitable for the classifier.
- Preparing the target variable representing triage acuity.
- Applying consistent preprocessing to training, testing and inference inputs.

Stage 3: ML model training

Different models will be trained and compared.

Possible models:

- Logistic Regression
- Random Forest

The dataset is divided into training and testing subsets. The training subset is used to fit the classifier, while the testing subset is used to estimate predictive performance.

Model	*Test accuracy*	*Balanced accuracy*
Random forest baseline	46.15%	39.34%
Logistic regression	51.92%	40.82%

The class distribution is imbalanced, particularly for acuity classes 1 and 4. Therefore, accuracy alone does not adequately describe the model's performance.

Output: A trained classifier and evaluation results that establish the predictive baseline before inference optimization.

Stage 4: Model conversion using ONNX

After training, the Logistic Regression classifier is exported into the Open Neural Network Exchange (ONNX) format.

ONNX provides a model representation that can be consumed by compatible inference runtimes. This allows the trained model to be deployed through OpenVINO without retraining it as a different classifier. The conversion stage prepares the model for optimized execution. The preprocessing pipeline must still supply inputs in the expected format, with compatible feature ordering, data types and encoding.

Stage 5: OpenVINO inference optimization

This module loads the converted ONNX model into Intel OpenVINO, compiles it for the CPU and executes inference.

The inference workflow can be represented as:

1. Load the converted model.
2. Compile the model for the CPU device.

3. Prepare an input batch or individual record.
4. Execute the inference request.
5. Obtain the predicted class.
6. Compare the predictions with those of the original scikit-learn classifier.

Stage 6: Performance evaluation

This module measures whether OpenVINO improves the execution efficiency of the trained classifier.

Two types of benchmarks are performed:

A. Classifier-only benchmark

This measures model execution time while excluding preprocessing and model compilation. The repeated benchmark screenshots report speed ratios of approximately 1.72× for batch inference and 1.83× for single-record inference in separate experiments.

B. End-to-end benchmark

This measures the complete inference pipeline, including preprocessing, while excluding OpenVINO model compilation.

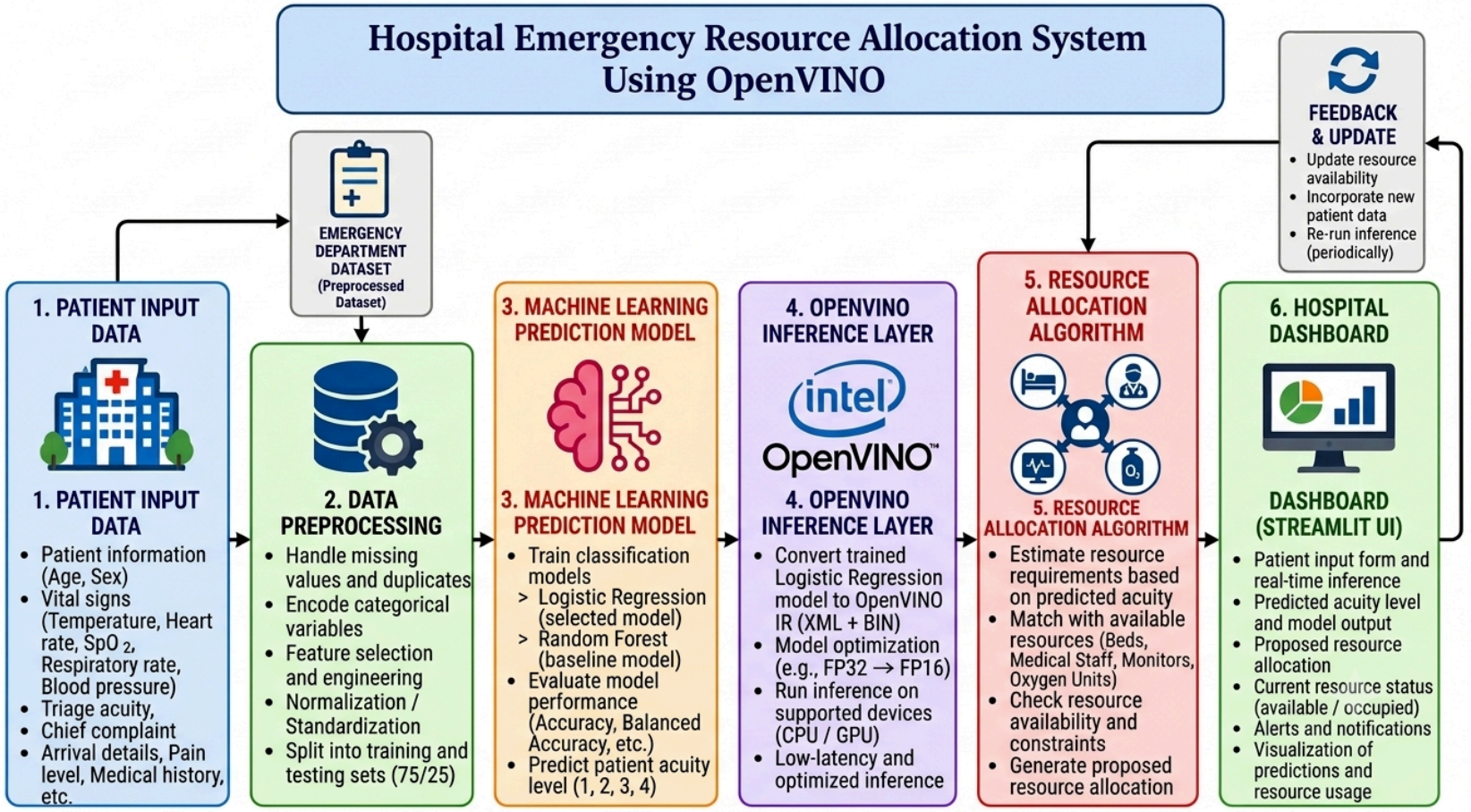
Run	***Scikit learn latency***	***Openvino pipeline mean latency***
1	***6.551330 ms***	***7.914014 ms***
2	***6.305257 ms***	***6.651685 ms***
3	***5.444239 ms***	***5.416421 ms***

Stage 7: Hospital resource inventory

This module demonstrates the representation of resource availability. In the current implementation, these inventory values are not shown to be automatically updated by predicted acuity, nor is an optimization algorithm shown to assign resources to individual patients.

$$R_{available} = R_{total} - R_{occupied}$$

9.Block Diagram of Proposed System:



10.Dataset Description and Required Software Installation Steps :

10.1.Primary Dataset: MIMIC-IV-ED

A particularly suitable dataset for the project is MIMIC-IV-ED, because it contains emergency-department data and has been used in recent research on emergency resource prediction.

The 2024 resource-allocation study discussed above used 391,472 ED visits from MIMIC-IV and predicted 144 resource-related targets.

Potential information available for modelling includes:

- Patient demographics
- ED arrival information
- Triage information
- Vital signs
- Medication information
- Diagnosis information
- Procedures
- Disposition
- Timing information
- ED stay information

It is a research dataset from a particular healthcare setting. Therefore, your report should describe the proposed system as a prototype decision-support system, not as a clinically validated production system.

10.2.Required Hardware:

- Intel-based laptop/PC
- Minimum 8 GB RAM recommended
- Sufficient storage for datasets
- Internet connection

Required Software:

Development Environment

- Windows
- Python 3.x
- VS Code
- GitHub

Python Libraries

- numpy
- pandas

- scikit-learn
- matplotlib
- openvino
- joblib
- flask

For optimization

- scipy
- pulp

Software Installation Steps

1. Install Python 3.x and verify: python --version

2. Create a virtual environment : python -m venv emergency_env

Activate it on Windows: emergency_env\Scripts\activate

3: Upgrade pip:python -m pip install --upgrade pip

4: Install ML libraries: pip install numpy pandas scikit-learn matplotlib

5: Install OpenVINO : pip install openvino

OpenVINO's current documentation provides Python installation and deployment guidance.

Verify installation :: python -c "import openvino; print(openvino.__version__)"

6: Install optimization library : pip install ortools / pip install pulp

7.Install Flask : pip install flask

11. Detailed description of the work carried out :

The Hospital Emergency Resource Allocation System Using OpenVINO was developed to study the application of machine learning in emergency department resource planning and to evaluate inference optimization using OpenVINO. The work includes dataset inspection, data preprocessing, machine learning model development, model conversion to ONNX, OpenVINO CPU inference, performance benchmarking, and resource availability analysis.

Module 1: Dataset Collection and Inspection

The first module involved examining the available emergency department dataset to understand its structure, identify relevant variables, and check data quality. The information included patient demographics, vital signs, triage information, visit details, diagnoses, medications, and procedure information, where available.

The dataset was inspected for missing values, class distribution, and the availability of information required for model training and evaluation.

Mathematical function for missing value percentage:

$$P_{missing} = \frac{N_{missing}}{N_{total}} \times 100$$

Where :
N_missing = number of missing values
N_total = Total number of values

This calculation helps identify incomplete features and determine whether data cleaning or missing-value handling is necessary.
Outcome: The dataset inspection provided an understanding of the available information and the data-quality issues to consider before model development.

Module 2: Data Preprocessing

Data preprocessing prepares the dataset for machine learning. The work involved examining missing and inconsistent values, identifying relevant features and the target variable, and preparing the data for model training and evaluation.

Numerical and categorical features may require different preprocessing techniques. The exact transformations depend on the operations implemented in the project code.

Mathematical function: Min–max normalization
When min–max normalization is applied, the original value is transformed as follows:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Here, (x) is the original value, (x_min) is the minimum value, and (x_max) is the maximum value of the feature.
This transformation scales values between 0 and 1 when the maximum and minimum values are different.

Mathematical function: Standardization

When standardization is applied, a feature value is transformed using:

$$z = \frac{x-\mu}{\sigma}$$

Where:

- x = Original value.
- μ = Mean.
- σ = Standard deviation.

It can help models such as Logistic Regression when input features have different numerical scales.

Module 3: Machine learning model development

The third module involved training and evaluating two classification algorithms: Random Forest and Logistic Regression. These models were used to investigate the prediction capability of the available emergency department data.

A. Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees. For classification, the final predicted class is commonly determined by majority voting among the trees.

The mathematical representation is:

$$\hat{y} = \{h_1(x), h_2(x), ..., h_T(x)\}$$

The model combines the outputs of multiple trees to determine the most frequently predicted class.

Result: The Random Forest model achieved an accuracy of 46.15% and a balanced accuracy of 39.34% in the reported experiment.

B. Logistic Regression Classifier

Logistic Regression is a supervised learning algorithm that estimates the probability of a class based on input features.

For binary classification, the sigmoid function is:

$$P(y = 1|x) = \frac{1}{1+e^{-z}}$$

Where:

$$z = w^T x + b$$

The sigmoid function converts the model score into a probability between 0 and 1.

For multiclass classification, the softmax function can be used:

$$P(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

This calculates the probability assigned to each class in a multiclass classification problem.

Result: Logistic Regression achieved an accuracy of 51.92% in the reported experiment.

C. Model Evaluation

The classification models were evaluated using accuracy and balanced accuracy.

Accuracy: Accuracy measures the proportion of predictions that are correct.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \times 100$$

Balanced Accuracy:

$$Balanced\ Accuracy = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k+FN_k} \times 100$$

This metric gives equal importance to each class.
Balanced accuracy is useful when the dataset has an unequal distribution of classes because it gives each class equal importance in the final metric.

Module 4: Model Conversion to ONNX

After training, the Logistic Regression model was converted into ONNX format. ONNX provides a standardized representation that enables compatible machine learning models to be used with supported inference runtimes.

Module 5: OpenVINO CPU Inference and Prediction Validation

The fifth module involved loading the ONNX model into the OpenVINO runtime and performing inference on the CPU. The predictions from OpenVINO were compared with the predictions generated by the original scikit-learn model.

Mathematical function: Prediction agreement

Prediction agreement measures the percentage of records for which both implementations produce the same predicted class.

$$A_{agreement} = \frac{N_{matching}}{N_{compared}} \times 100$$

Module 6: Performance Benchmarking

This module evaluated the inference performance of the scikit-learn model and the OpenVINO implementation. Two types of benchmarking were considered: classifier-only inference time and end-to-end pipeline latency.

A. Classifier-only inference time

The reported classifier-only benchmark for a batch of 207 records produced the following measurements:

- Scikit-learn average time per batch: 0.220828 ms.
- OpenVINO average time per batch: 0.106526 ms.

Mathematical function: Speedup

Speedup measures how many times faster the baseline execution time is relative to the optimized execution time.

$$S = \frac{T_{baseline}}{T_{OpenVINO}}$$

B. End-to-end pipeline latency

End-to-end latency measures the execution time of the broader prediction pipeline, potentially including preprocessing and other data-handling operations in addition to model inference.

$$\overline{T} = \frac{1}{n}\sum_{i=1}^{n} T_i$$

Module 7: Hospital Resource Availability Analysis

The resource availability module examined the number of available emergency beds, medical staff, patient monitors, and oxygen units. The total and occupied quantities were used to calculate the remaining available resources.

Mathematical function:

Resource availability percentage:

$$R_{available} = R_{total} - R_{occupied}$$

13. Performance Parameters Considered:

The performance of the Hospital Emergency Resource Allocation System Using OpenVINO is evaluated using parameters related to machine learning prediction quality, inference efficiency, model consistency, and resource availability. These parameters help assess whether OpenVINO can improve inference performance while maintaining consistency with the original machine learning model.

Inference Latency

Inference latency is the time required by a model to generate predictions for a given input. Lower latency is desirable in emergency healthcare applications because timely predictions can support faster resource-planning decisions.

Formula:

```
Inference Latency = End Time − Start Time
```

In the classifier-only benchmark, the scikit-learn model recorded an average inference time of 0.220828 ms per batch, while OpenVINO CPU inference recorded 0.106526 ms per batch.

Inference Speedup

Inference speedup measures the performance improvement obtained by executing the model using OpenVINO compared

with the baseline implementation.

Formula:
Speedup = Baseline Inference Time / OpenVINO Inference Time

Using the measured classifier-only benchmark:

`Speedup = 0.220828 / 0.106526 ≈ 2.07×`

The result indicates approximately 2.07 times faster classifier-only inference with OpenVINO CPU compared with the measured scikit-learn baseline.

Prediction Accuracy

Prediction accuracy measures the proportion of correctly classified records among all evaluated records. It is used to assess the classification performance of the machine learning model.
Formula:

Accuracy = Number of Correct Predictions / Total Number of Predictions

The experimental results were:

- Random Forest accuracy: 46.15%.
- Logistic Regression accuracy: 51.92%.

Logistic Regression achieved higher accuracy than the Random Forest baseline in the reported experiments. These results represent performance on the evaluated dataset and do not establish clinical effectiveness.

Balanced Accuracy

Balanced accuracy accounts for the performance across different classes, making it useful when the dataset contains unequal numbers of examples in each class.

Formula:

Balanced Accuracy = Average Recall Across All Classes

The Random Forest model achieved a balanced accuracy of 39.34%. This indicates that the model's classification performance across the different classes requires further improvement.

14.Implementation Details-CPU & GPU

14.1.CPU Implementation

The system uses Python, Scikit-learn, and OpenVINO to predict patient triage acuity using a trained Logistic Regression model. The CPU executes model inference using patient vital signs and demographic information. The predicted acuity and class probabilities are displayed on the dashboard, and a rule-based module proposes emergency beds, medical staff, patient monitors, and oxygen units. The CPU benchmark, conducted over ten repetitions of 1,000 inference calls after 50 warm-up calls, recorded a mean latency of 0.168683 ms and a median latency of 0.167199 ms. These results indicate that CPU execution provides lower inference latency for the current single-input workload.

14.2.GPU Implementation

The model was successfully compiled and executed on GPU.0 using the OpenVINO GPU plugin. A technical test using an input tensor of shape (1, 168) confirmed successful GPU inference. In the repeated benchmark, GPU inference recorded a mean latency of 0.543031 ms and a median latency of 0.391393 ms. The CPU was approximately 3.22 times faster than the GPU based on their mean latencies. This suggests that GPU execution overhead may outweigh the benefits of parallel processing for the current small Logistic Regression model. Therefore, CPU execution is preferred for the tested workload, while GPU support has been demonstrated successfully.

Parameter	CPU	GPU.0
Model execution	Successful	Successful
Mean latency	0.168683 ms	0.443031 ms
Minimum latency	0.123318 ms	0.391393 ms
Maximum latency	0.261083 ms	0.90024 ms
Standard deviation	0.038313 ms	0.251192 ms
Timed repetitions	10	10
Timed calls per repetition	1000	1000

15. RESULTS:

The experimental results demonstrate the performance of the proposed hospital emergency resource allocation system in patient acuity classification, inference optimization, and resource availability monitoring. Logistic Regression achieved an accuracy of approximately 51.92%, outperforming the Random Forest baseline accuracy of 46.15% on the test dataset. The OpenVINO benchmark showed lower average inference latency on the CPU than on the tested GPU device, while the resource availability visualization provides an overview of the hospital resources available for emergency planning.

Graphs:

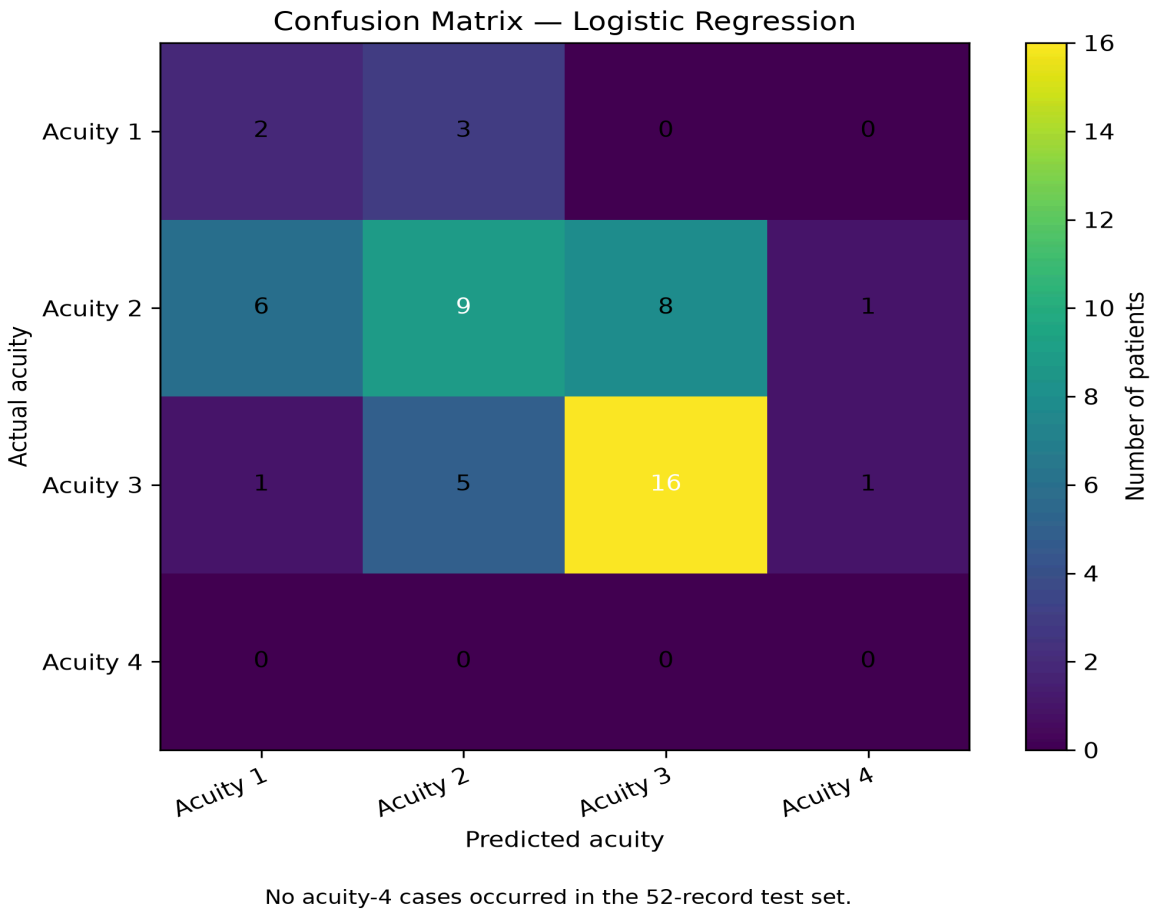


Figure 1:Confusion matrix

The confusion matrix illustrates the classification performance of the Logistic Regression model by comparing the actual patient acuity levels with the predicted acuity levels. It shows the number of correct and incorrect predictions for each class, helping identify the model's classification strengths and errors.

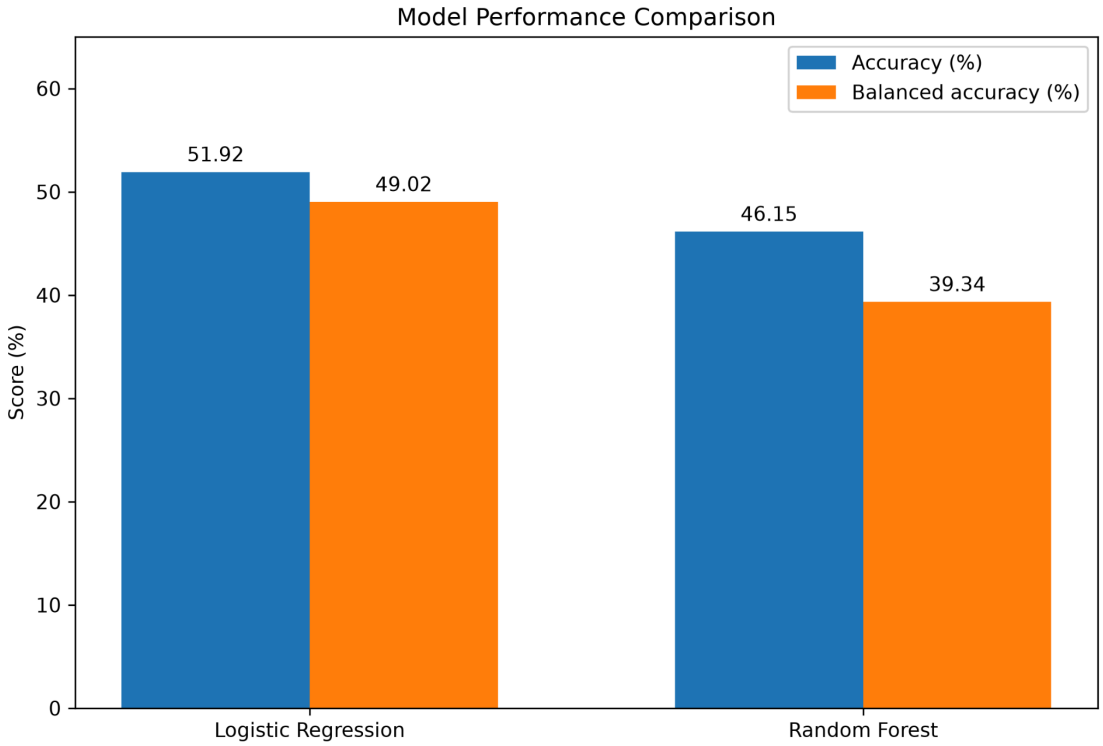


Figure 2:model performance

This graph compares the performance of Logistic Regression and Random Forest models using accuracy and balanced accuracy metrics. The results show that Logistic Regression achieved higher accuracy and balanced accuracy on the test dataset than the Random Forest baseline model.

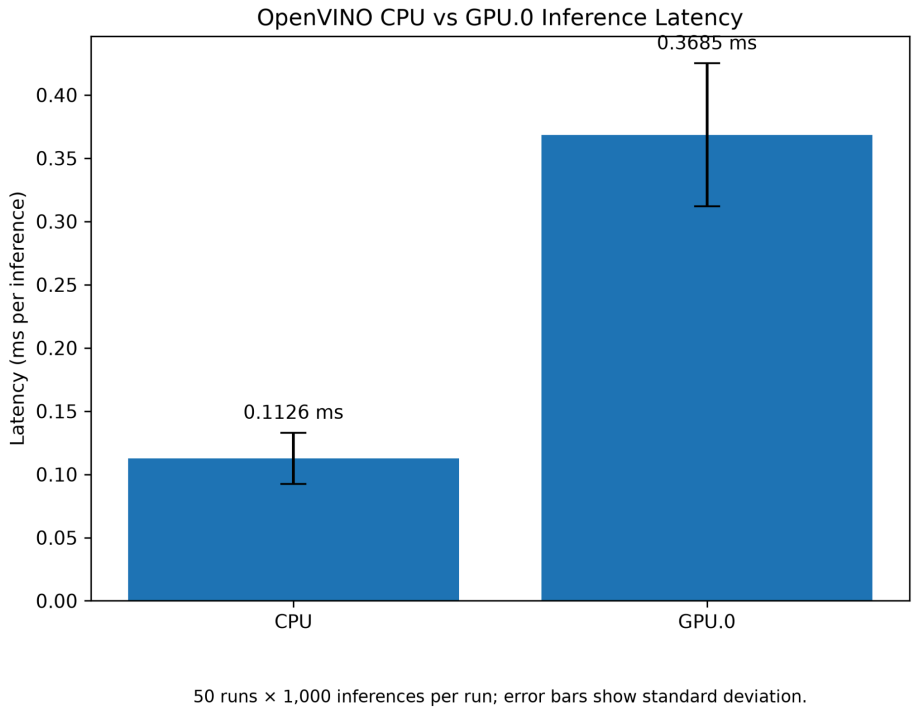


Figure 3:cpu vs gpu latency

This graph compares the average inference latency of the OpenVINO model when executed on CPU and GPU devices. In the measured benchmark, CPU inference was faster than GPU inference, indicating that CPU execution was more efficient for this model and workload under the tested conditions.

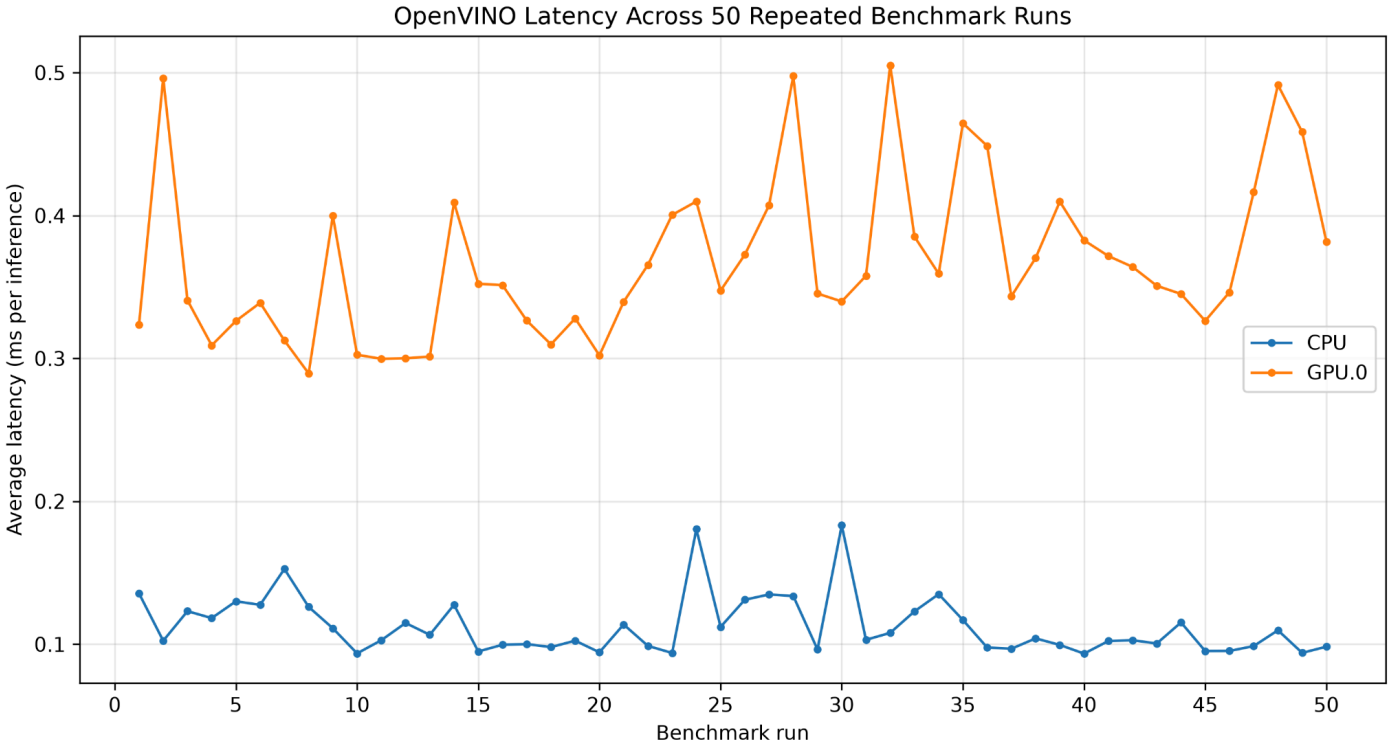


Figure 4:repeated run latency

This graph presents the inference latency recorded across 50 repeated benchmark runs for CPU and GPU execution using the OpenVINO model. It highlights variations in execution time and demonstrates the consistency of inference performance across repeated measurements.

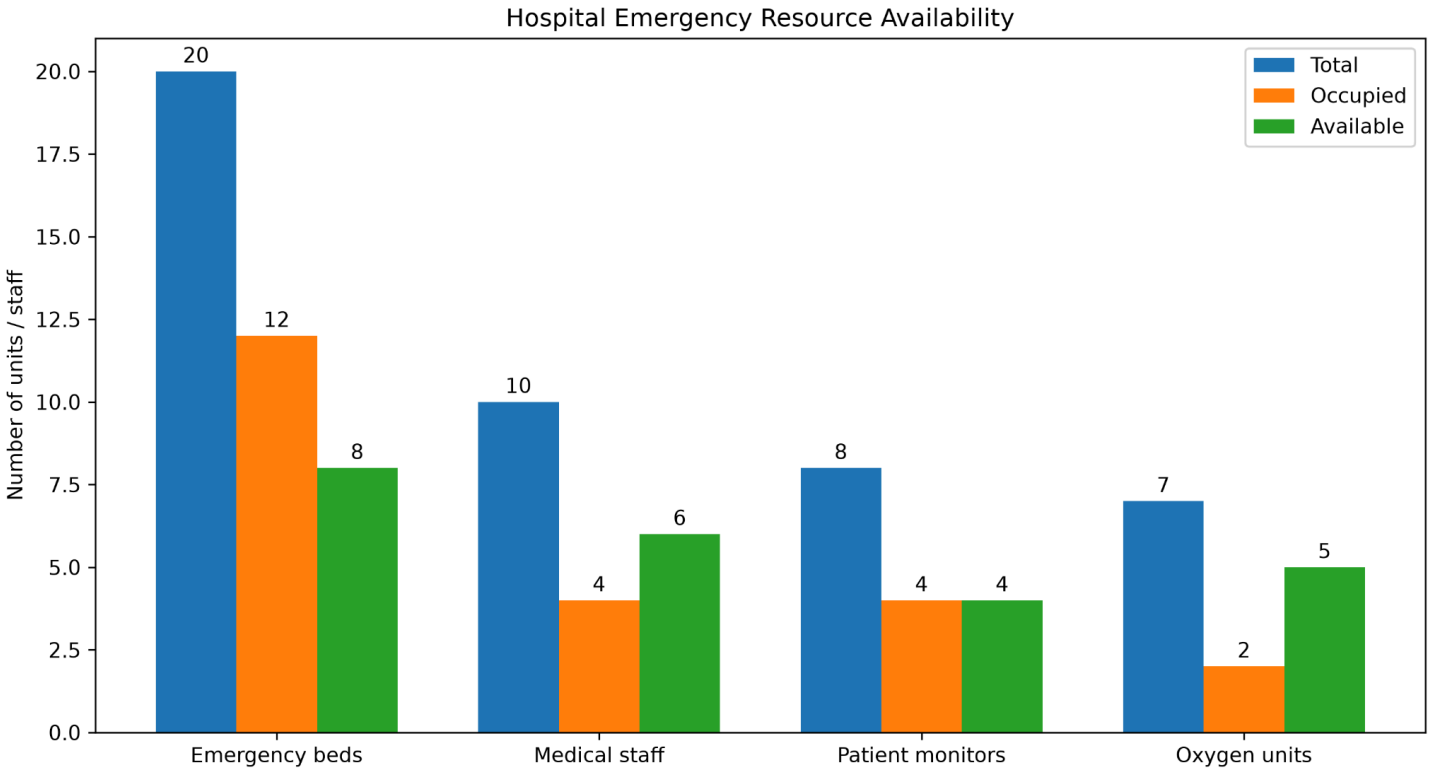


Figure 5:resource availability

This graph illustrates the total, occupied, and available quantities of essential hospital resources, including beds, staff, monitors, and oxygen units. It provides a visual overview of resource utilization and availability to support emergency resource planning and allocation.

Resource Allocation and Decision-Making

The system uses the machine learning model to predict patient acuity based on the available patient information. The predicted acuity, together with the current resource inventory, can support prioritization and resource planning by helping identify emergency cases and available hospital resources. The system is intended as a decision-support prototype, and all patient prioritization and resource allocation decisions require verification by qualified healthcare personnel.

OpenVINO Inference Optimization

OpenVINO was used to load and execute the converted machine learning model for inference benchmarking on supported hardware devices. CPU and GPU inference latencies were measured over repeated runs to compare execution performance. In the tested configuration, the CPU achieved lower average inference latency than the GPU, suggesting that CPU execution was more suitable for this particular model and benchmark workload.

Limitations of the System

The current system has limitations arising from the size and class distribution of the dataset, which may affect the reliability of patient acuity predictions. In the test dataset, no samples belonged to acuity class 4, so the model's performance for that class could not be evaluated. In addition, the inference benchmark measures model execution under a controlled input rather than the complete end-to-end hospital workflow. Further validation using larger and more representative datasets is required before considering real-world clinical use.

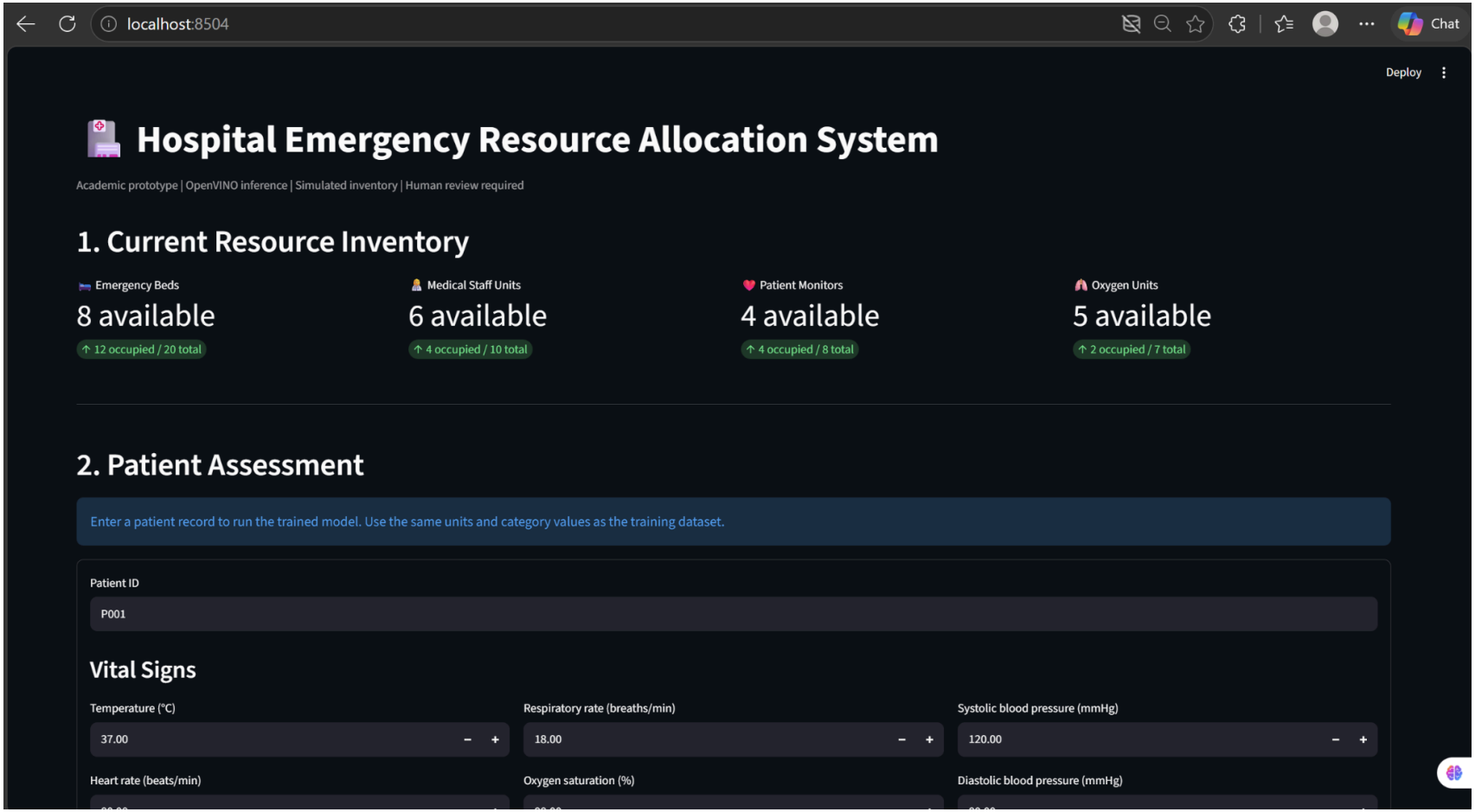


Figure 6:Hospital Emergency Resource Allocation System UI

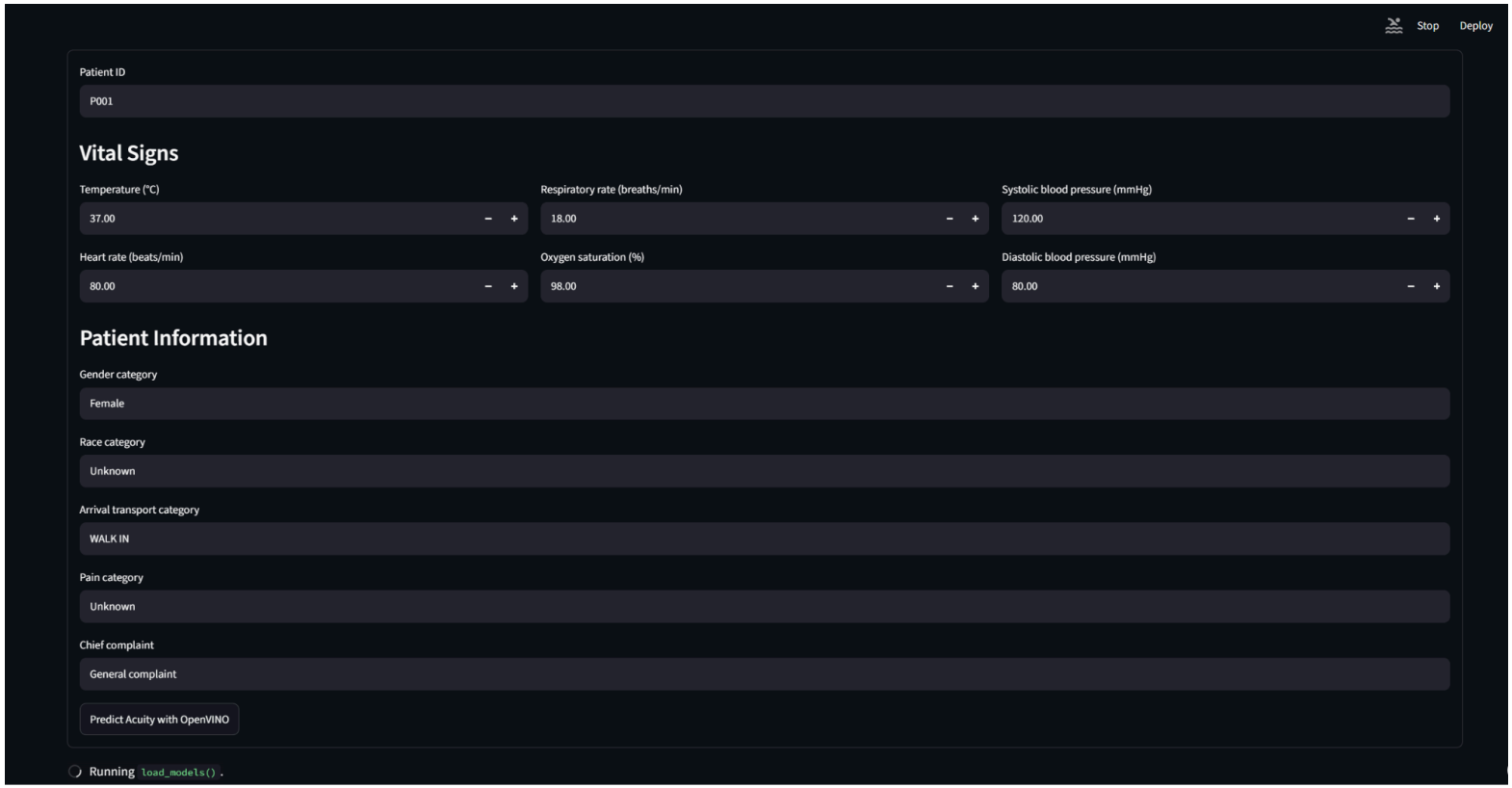


Figure 7: Patient assessment

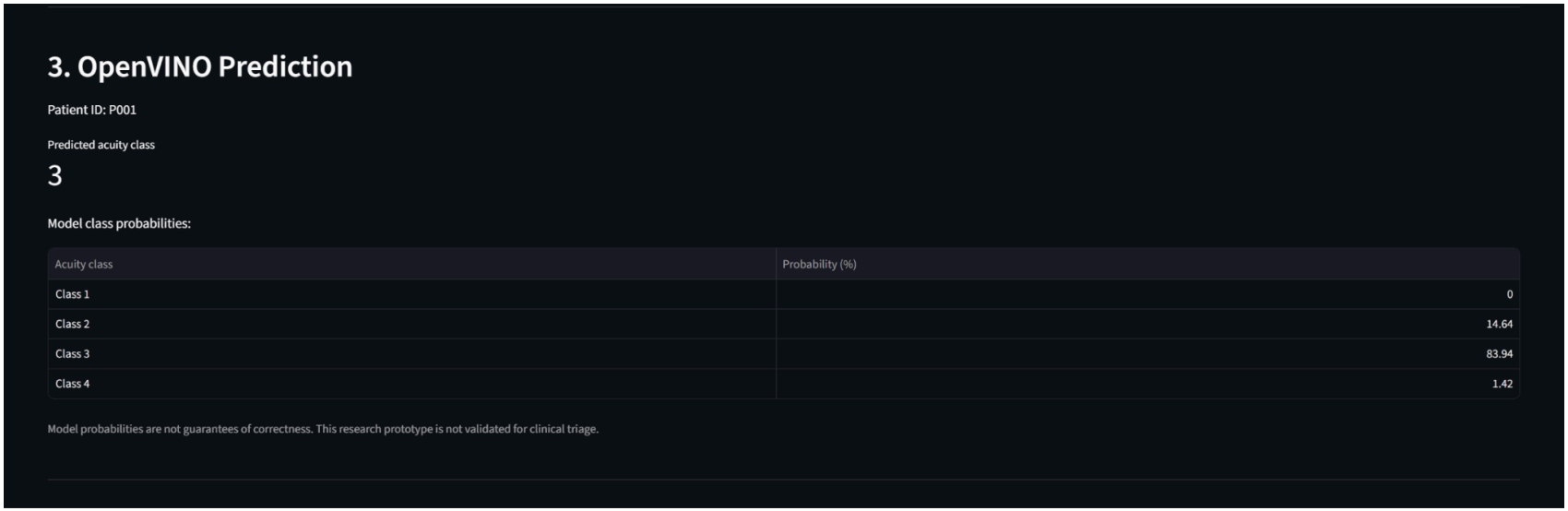


Figure 8:OpenVINO Inference Performance UI



Figure 9:Resource allocation proposal

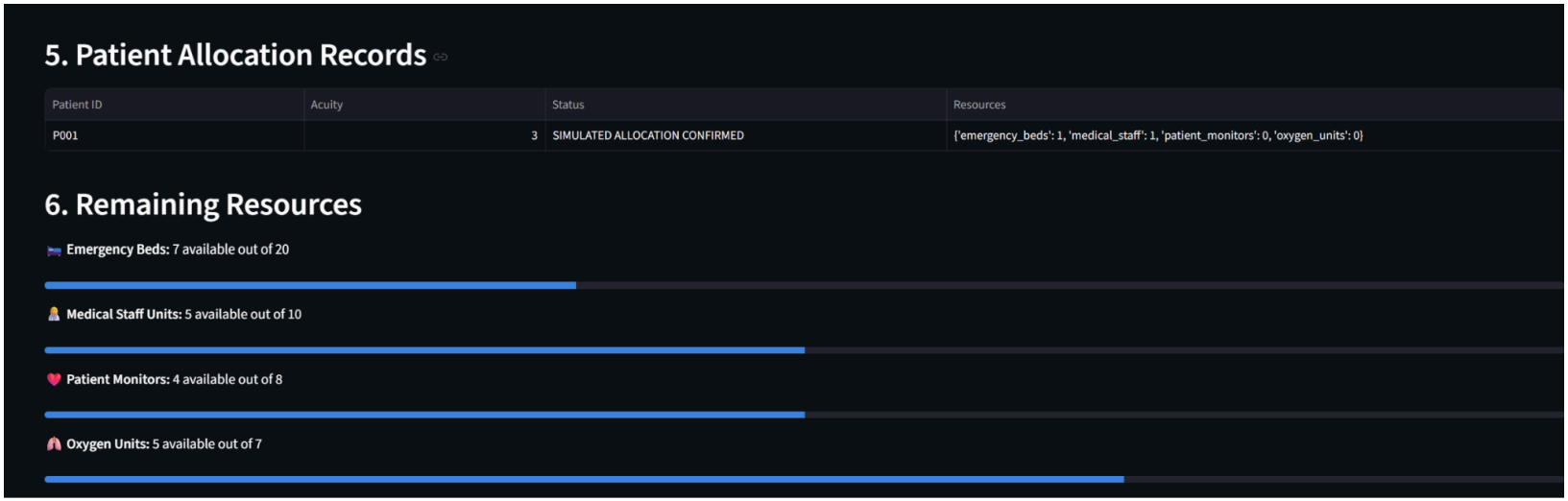


Figure 10: Records and remaining records

16.Conclusion:

This project presents a prototype hospital emergency resource allocation system that combines machine learning-based patient acuity classification, OpenVINO inference, and hospital resource availability monitoring. The experimental evaluation compares two classification models and measures CPU and GPU inference latency, while resource visualization supports an understanding of available emergency resources. Although the results demonstrate the feasibility of integrating these components into a single prototype, improvements in model accuracy, dataset representation, and system-level validation are needed before practical clinical deployment.

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